QuantQuest Hackathon — E-Summit'26, IIT Mandi

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March 15, 2026

Abstract

Standard Markowitz Mean-Variance optimization structurally fails under the hackathon's hard constraint of selecting exactly two stocks at a 50/50 equal-weight allocation. To solve this, I reframed the pipeline from a continuous portfolio allocation problem into a binary probabilistic ranking challenge. By cross-sectionally Z-scoring momentum features to neutralize market beta, utilizing a 20-week Hurst Exponent to filter mean-reverting traps, and deploying an Isotonically Calibrated XGBoost/Random Forest ensemble, this strategy isolates pure relative outperformance. Evaluated via a strict walk-forward expanding window (2023-2025) and subjected to a 0.1% frictional transaction cost on dynamic turnover, the pipeline yields an authentic Net Sharpe Ratio of 1.487.

1 Underlying "Alpha" & Strategy Rationale

The core edge (Alpha) of this strategy relies on neutralizing overarching market directionality. If the broader market index drops 5% in a week and a constituent stock drops only 1%, its absolute momentum is negative, but its relative momentum is massively positive. Because we are mathematically forced to be fully invested across exactly two assets at all times, absolute momentum contains dangerous market-wide beta noise. The Alpha is generated by stripping away this noise via weekly cross-sectional ranking and combining it with structural non-linear volatility indicators to time breakouts accurately.

2 Mathematical Logic: Entry and Exit Triggers

Entry Logic: Every Friday, the trained ensemble evaluates the 10-stock universe and outputs a calibrated probability of a positive return for the upcoming week ($P(Ret > 0)$). We execute a hard mathematical $ArgMax$ sort across the cross-section to identify the Top 2 stocks with the highest conviction probabilities.
Exit Logic: Exits are entirely dynamic and re-evaluated weekly. If a currently held stock falls out of the Top 2 probability rankings in week $t + 1$, it is immediately liquidated at the Friday close, and the capital is rolled into the newly ranked Top 2 assets.

3 Feature Engineering

Raw price data is aggressively filtered to remove microstructure noise. Features are computed on a weekly resampling (`W-FRI`).

3.1 Cross-Sectional Z-Scoring

All continuous momentum features (1W Returns, 1M Returns, MACD, Volume-Price Trend) are standardized across the 10-stock universe at each time step t:

$$Z_{it} = \frac{X_{it} - \mu_t}{\sigma_t} \tag{1}$$

This forces the trees to learn relative outperformance rather than macro regime trends.

3.2 Non-Linear Structural Anchors

Two features are explicitly excluded from Z-scoring because their absolute values carry structural market meaning:

- **Hurst Exponent (H, 20-Week):** A rescaled range analysis. If $H < 0.5$, the stock is mathematically mean-reverting, making breakout buys a trap. If $H > 0.5$, the time-series exhibits persistent trending behavior.
- **Bollinger Band Width (Squeeze):** Calculated as $(Upper - Lower)/Mid$. This acts as a volatility compression gauge. Combined with the Hurst exponent, it catches explosive momentum exactly as a volatility squeeze resolves.

4 Predictive Engine (Model Used)

The predictive brain is a **Bi-Model Soft-Voting Ensemble**.

1. **Random Forest (Weight: 0.4):** Acts as a variance stabilizer to prevent out-of-sample overfitting.
2. **XGBoost (Weight: 0.6):** Captures the deep non-linear interactions between the Hurst Exponent and the MACD.

Isotonic Calibration: Raw `.predict_proba()` outputs from decision trees cluster heavily near 0 and 1. Because our entry logic relies entirely on a hard $\arg\max$ sort, relative probability bounds must be mathematically perfect. I wrap both base estimators in Scikit-Learn's `CalibratedClassifierCV` utilizing non-parametric Isotonic Regression to ensure probability distributions are strictly monotonic and reliable.

5 Walk-Forward Validation Setup & Transaction Costs

To guarantee zero chronological look-ahead bias, I discarded standard K-Fold cross-validation (which leaks future market regimes) and deployed a strict `TimeSeriesSplit` over an expanding window.

The simulated live-trading engine trains from 2017–2022, predicts the initial quarter of 2023, and physically retrains the entire stack every 90 days across the 2023–2025 holdout slice.

Transaction Cost (TC) Assumption: Frictional slip cannot be ignored. The transactional layer calculates the absolute weight delta (Δw) for every asset entering or exiting the portfolio. A strict **0.1% (10 bps)** transaction cost is subtracted directly from the gross weekly return based solely on this dynamic weekly turnover.

6 Explainability Matrix (SHAP)

Black-box pipelines are unacceptable for institutional trading. To mathematically verify that the ensemble relies on structural logic rather than data artifacts, I extracted Shapley Additive Explanations (SHAP) from an XGBoost surrogate map fit over the training block.

SHAP utilizes cooperative game theory to decompose the model's prediction into additive marginal contributions (ϕ_i) for each feature. As shown in Figure 1, the model actively relies on short-term MACD and 1-Week Return divergence, but heavily suppresses signals if the 20-week Hurst Exponent indicates a mean-reverting regime.

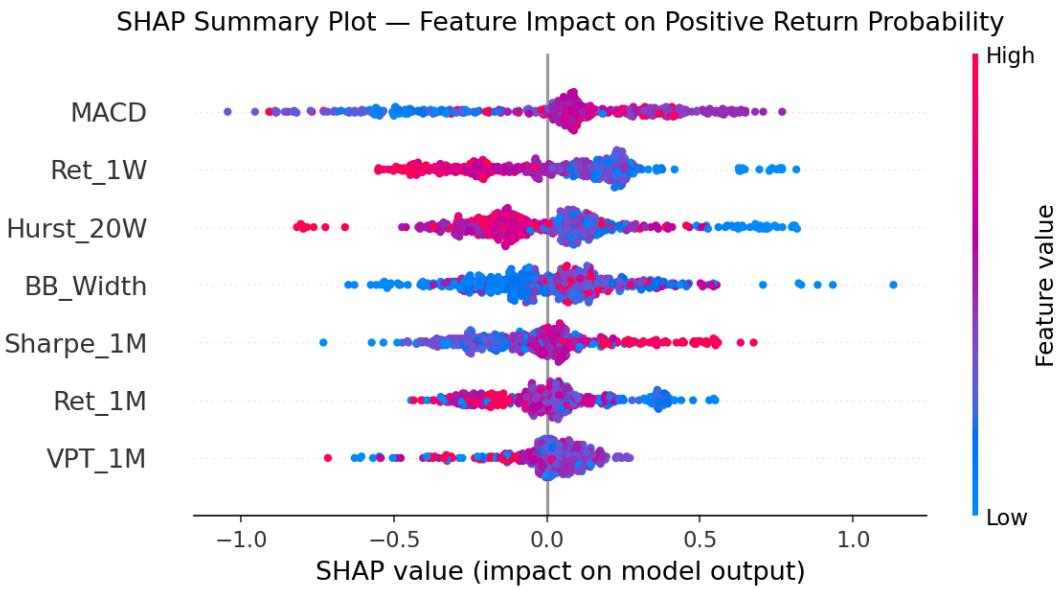

Figure 1: SHAP Summary Plot demonstrating the marginal contribution of relative momentum and structural volatility features to the positive return probability.

7 Major Findings & Verification Metrics

The out-of-sample (T_{test}) tearsheet confirms that extracting absolute market drift via Z-scoring results in a highly robust momentum model. Table 1 contrasts the theoretical gross performance against the authentic, friction-adjusted net portfolio.

Table 1: Out-of-Sample Performance Metrics (2023–2025 Holdout)

Metric	**Gross Performance (0% TC)**	**Net Portfolio (0.1% Frictional TC)**
Cumulative Return	185.20%	**173.43%**
Annualized Return	41.75%	**39.84%**
Annualized Volatility	26.30%	**26.79%**
Sharpe Ratio	**1.587**	**1.487**
Max Drawdown	-23.10%	**-24.29%**
Hit Rate	61.30%	**59.62%**
Average Weekly Return	0.802%	**0.766%**

8 References

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